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To cite this article: Greta de Groat (2015) A History of Video Game Cataloging in U.S. Libraries, Cataloging & Classification Quarterly, 53:2, 135-156, DOI: [10.1080/01639374.2014.954297](https://doi.org/10.1080/01639374.2014.954297)

To link to this article: <https://doi.org/10.1080/01639374.2014.954297>



Published online: 22 Dec 2014.



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A History of Video Game Cataloging in U.S. Libraries

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Cataloging practices for video games have been in flux since the late 1970s, often lagging behind technological developments. This article describes the history of descriptive cataloging practices and MARC coding for video games. Also discussed are the special problems presented by Library of Congress subject heading and uniform title practice, which were developed for book cataloging rather than for the cataloging of video games themselves.

KEYWORDS *descriptive cataloging, subject cataloging, video games, Anglo-American Cataloguing Rules (AACR2), MARC 21 formats, Resource Description and Access (RDA), Library of Congress Subject Headings (LCSH)*

As relatively new materials to library collections, video games have been an awkward fit into the regular library cataloging workflow. As games developed and non-book materials became more integrated into library cataloging, the cataloging rules and practices have struggled to keep up and to find adequate ways of representing these materials. There have never been specific rules for video games. Descriptive cataloging has usually fallen under the more general rules for computer-related materials which have been in constant flux. Subject/genre practices are based on practice for books, often with unfortunate results. These issues have led to confusion and inconsistency in catalog records.

Descriptive cataloging rules in the United States are governed by an international committee of representatives from the English-speaking cataloging world, formerly the Joint Steering Committee for the Revision of AACR (now the Joint Steering Committee for Development of RDA). The domain

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Received June 2014; revised August 2014; accepted August 2014.

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of descriptive cataloging rules is the description of the resource in physical terms, the mode of its usage, and names of creators and other associated persons or corporate bodies. These are refined and extended by the Library of Congress, who publishes their interpretations of the rules and their practices and keeps a list of “authorized” names—the form of names to be used in catalog entries. These practices are followed by most libraries in the United States. The Library of Congress controls the domain of the subject and genre. These are governed by Library of Congress practices and are not based on the descriptive cataloging rules. Finally, all of this data must be encoded in a data structure for dissemination among libraries and use in online catalogs. The standard in the United States is MARC 21, formerly US MARC. This data structure, created in the late 1960s for the purpose of printing catalog cards, divides data into fields, subfields, and byte strings for machine manipulation. Some of the fields were designed for indexing, some were not. All of the data called for in the descriptive and subject cataloging rules must be encoded (sometimes shoehorned) into one or more MARC fields.

DESCRIPTIVE CATALOGING RULES

Neither the ALA Cataloging Rules (1941)¹ nor Anglo-American Cataloging Rules, North American Edition (1967)² mention computer materials. Chapter 12 Revised: Audio Visual Media and Special Instruction Materials, published separately in 1975 after consultations between LC and the Joint Advisory Committee on Nonbook Materials, gave guidelines for many types of audio visual materials, but stated: “Only machine-readable data files were deleted as a category from the original list of media; the authors feel that this material requires further study before cataloging rules can be properly formulated.”³

The first appearance of any guidelines is in *Anglo-American Cataloguing Rules*, Second Edition (AACR2; 1978), as Chapter 9, machine-readable data files. These are defined as: “a body of information coded by methods that require the use of a machine (typically a computer) for processing. . . . The term machine readable data file embraces both the data stored in machine-readable form and the programs used to process that data.”⁴

This first AACR2 Chapter 9 is not lengthy, but does give a few specific instructions applicable to machine readable data files.

The “chief source of information (9.0B1),” the place that a cataloger is instructed to use as the source of the title and much other information, is specified as the “internal user label,” when the resource has one. Rule 9.1C allows the optional addition of a “general material designation” [machine-readable data file] following the title. Despite its optional status, the general material designation (or GMD) has generally been used by U.S. libraries for nonprint materials. As with other AACR2 rules for edition, 9.2B3, allows catalogers to “supply a suitable brief statement . . . and enclose in square

brackets” when a file lacks an edition statement but is known to contain “significant changes from previous editions.” Rule 9.2B4 instructs catalogers to not consider minor changes or a change in physical form to be a new edition. This seems rather startling at first until one consults the File Description Area (9.5), which is unique to this chapter. The .5 designation in other chapters refers to the extent and physical description, but in chapter 9 it referred to data files without reference to physical media. Accompanying material (9.5D) may be recorded, whether it is textual (e.g., a codebook) or is another program. This results in statements such as:

1 data file (3,000 logical records) + 1 program file
 1 program file (300 statements, COBOL) + 1 data file (1,000 logical records)
 1 data file (30,000 logical records) + 1 codebook (24 p. ; 25 cm.) + 1
 codebook (990 logical records)

Video games already existed by the time that the Chapter 9 rules were first published, but I have been unable to find any examples cataloged according to this first iteration of Chapter 9 prior to 1983.

The first manual dedicated to computer materials was Sue A. Dodd’s *Cataloging Machine-Readable Data Files: An Interpretive Manual*, 1982, which only mentions video games in passing.⁵ In 1983, Nancy Olson produced *A Manual of AACR2 Examples for Microcomputer Software and Video Games*, based on the 1978 AACR2 rules. At that time, the American Library Association Committee on Cataloging: Description and Access (CC:DA) had a task force preparing guidelines for cataloging computer software, and Olson’s *Manual* records the task force recommendation as “the projected guidelines also cover electronic toys or games that are typically issued in cartridge carriers and are manipulated by hand controls. These latter materials are technically under Chapter 10, and a GMD from chapter 10 should probably be used. . .”⁶ This means they recommended video games be treated as realia. The task force also advocated a compromise with the 1978 rules, adding to the program statement rules in area 9.5 an indication that the files were on some physical medium,⁷ with resulting examples such as:

program files in 1 computer cartridge (Atari 2066)

The examples in the *Manual* show games cataloged with a mix of practices; having the GMD [game] from Chapter 10 and the program files statement from Chapter 9—while omitting the actual number of program files as they were impossible for a cataloger to ascertain.⁸ However by the time that the ALA published *Guidelines for Using AACR2 Chapter 9 for Cataloging Microcomputer Software* in 1984, the scope includes the statement: “These guidelines apply to electronic games of the kind typically issued in cartridge carriers and manipulated by hand controls.”⁹ A System requirements note is explicitly advocated in the American Library Association (ALA) *Guidelines*, along with a Disk characteristics note.¹⁰

Examples show games cataloged purely according to Chapter 9. The ALA *Guidelines* also codify the addition of the physical medium, but with the programming language and/or computer name attached to the program statement rather than the statement of medium,¹¹ as in:

- 1 program file (MBASIC, C-BASIC) on 1 computer disk
- 1 program file (PASCAL, Apple IIe) on 1 computer disk

1985's *Cataloging Microcomputer Files: A Manual of Interpretation for AACR2* by Dodd and Ann Sandberg-Fox noted that the software field was evolving so quickly that rules were becoming rapidly outdated.¹² They also offered an explanation of another recommendation in the ALA *Guidelines* that had far-reaching consequences in cataloging of computer materials—to record all technical information, such as make and model of machine, memory, operating system, and so on in a single Systems requirement note.¹³ This was so that the user did not have to look at more than one place within the catalog record for the technical information. This made sense at a time when catalog records were not long out of the card environment and were searchable only by title, author, and subject. In this environment, it was reasonable to optimize the catalog record for human eyes, since the possibility of machine manipulation of the data was very limited and the computer field's rapid evolution made it difficult to standardize vocabulary that would be useful in indexing.

The 1988 revision of AACR2 documents several changes in practice. The most obvious change is that Chapter 9 was now called "Computer files." The definition from 1978 has been replaced by the simpler statement: "The rules in this chapter cover the description of files that are encoded for manipulation by computer. These files comprise data and programs. Computer files may be stored on, or contained in, carriers available for direct access or remote access."¹⁴ Electronic devices themselves or programs residing in their memory would be cataloged according to Chapter 10 as three-dimensional artefacts or realia. The instruction for the chief source of information from 9.0B1 has evolved from taking the title from an "internal user label" to using the "title screen," and, when lacking, from the physical carrier or labels, and so on. There is a footnote, however, to note that the title screen may not be "available" because of the cataloger's lack of access to equipment to mount or read the file. For practical purposes, this means that if a cataloger does not mount the file, they are to take the title from the label. This has implications for video games, as many catalogers do not routinely run the game and generally do not have the equipment available to do so. With this in mind, in the 1988 revision it was specified in rule 9.7B3 to always give a source of title note for computer file materials. In addition, in keeping with the change in the chapter title, the GMD was changed to [computer file].

A new data element, File characteristics, was added in the 1988 rules (rule 9.3, specifically excluded in the 1978 rules). This area incorporates data that in the 1978 rules was put in the “file description area” in 9.5, which corresponded to the physical description area for other types of materials. The terms computer data, computer program(s) or computer data, and program(s) are specified as the vocabulary to be used, in addition to parenthetical information on files, records, and statements, when known.

The rapid development in commercial computer carriers made it necessary for a re-thinking of 9.5 physical description, which is why the file data was moved to 9.3. In its place in 9.5 are instructions which more accurately parallel the descriptive practices for other materials: number and type of physical units, sound and color, and dimensions. There is a list of general terms for physical units and a list of more specific terms, which are the ones which libraries have generally used. These are computer chip cartridge, computer tape cartridge, computer tape reel, computer laser optical disk, computer laser optical card. One is generally to give dimensions in inches, except for computer cards, which are given in centimeters, or reels, which are given no dimensions. Here are examples of some resulting physical description formulas:

- 1 computer chip cartridge : sd., col. ; 3 $\frac{1}{2}$ in. + 1 user's guide
- 2 computer tape reels : 6,150 bpi + 1 codebook
- 1 computer laser optical disk : col. ; 4 $\frac{3}{4}$ in.

In the Notes area, rule 9.7B1 mandated the System requirements note, which had been recommended in the *ALA Guidelines* of 1984. Catalogers are instructed to make the note based on information readily available and to begin the note with the wording “System requirements,” providing, in this order, the make and model of the computer on which the resource is to run, the amount of memory, operating system, software requirements (including programming language), and the kind and characteristics of required or recommended peripherals. Presumably this information would include the video game platform, although this is not explicitly stated. There are no instructions for a Disk characteristics note.

9.7B1 also instructs the cataloger to note the “nature and scope of the file unless it is apparent from the rest of the description” and gives the example “Game.” However, apparently most catalogers have considered the nature of the resource “apparent” and do not give this note.

The 1988 version of AACR2 was amended in 1993 and 1999, with a wholesale replacement of Chapter 9 in 2001. Computer resources had been continuing to evolve and there had been notable development of the Internet and World Wide Web, which needed accommodation in the rules. However, as of this date, few games on the web have been cataloged, despite the fact that this is an increasingly common mode of distribution.

Another, more temporary development, was the appearance of guidelines for interactive multimedia. This was defined as: “media residing in one or more physical carriers . . . or on computer networks. Interactive multimedia must exhibit both of these characteristics: (1) user-controlled, nonlinear navigation using computer technology and (2) the combination of two or more media (audio, text, graphics, images, animation, and video) that the user manipulates to control the order and/or nature of the presentation.”¹⁵ These seem like relatively ordinary characteristics for computer resources, but at the time they caused consternation because of the combination of types of content and sometimes physical carriers. The new guidelines quickly fell into obsolescence as CD-ROMs and DVD-ROMs became mainstream. The term “interactive multimedia” still hangs on in various places in subject/genre cataloging practice and in the MARC 21 format. The GMD [interactive multimedia] was approved for these materials and is still occasionally encountered in catalog records for computer games.

The title of the new Chapter 9 in the AACR2 2001 amendments was changed to Electronic Resources, and the GMD was duly changed to [electronic resource].¹⁶ The rule for 9.0B1 for choice of the chief source of information became “the resource itself.” The prescribed vocabulary for 9.3 was changed to “electronic data,” “electronic program(s),” or “electronic data and program(s).” Examples throughout the chapter were updated, though many archaic ones remained. The names of the physical carriers in 9.5B1 were tweaked: computer disk was retained for diskettes, but for CD-ROMs, computer laser optical disk became computer optical disc. Catalogers were also given the option of using “conventional terminology” to represent the carrier; the examples given were:

- 1 CD-ROM
- 2 photo CDs
- 1 DVD.

Many catalogers began exercising this option, as did the Library of Congress, although no instructions were given for when a cataloger was faced with recordable discs, such as CD-R or DVD-R as to whether that should be recorded in this field or in a note. Meanwhile, many catalogers continued to use the wording given in 9.5B1.

- 1 computer disk : col. ; 3 $\frac{1}{2}$ in.
- 1 computer optical disc : sd., col. ; 4 $\frac{3}{4}$ in.

This history of constantly changing terminology, plus the “conventional terminology” option with no prescribed vocabulary has led to a wide diversity of physical description formulations for computer games, particularly CD-ROM and DVD-ROM types. This makes it very difficult to consistently identify particular classes of discs in an automated manner.

Another major revision of AACR2 came out in 2002 with a replacement of all the text.¹⁷ This was updated in 2003, 2004, and 2005. Examples were again updated to some extent, but the major changes here were the complete elimination of 9.3 and more tweaking of the physical description area in 9.5B, which generally did not apply to game cataloging.

Resource Description and Access, known as RDA, is the successor to AACR2, and is slowly being adopted across the English speaking world and in many other countries as well.¹⁸ While it has been reconfigured and given a more rigorous intellectual underpinning and most data elements have been split into separate rules, it is still closely allied with AACR2 and in the commonly used implementation still uses many of the same formulations and punctuation (AACR2 uses International Standard for Bibliographic Description [ISBD] punctuation).¹⁹

The major differences between RDA and AACR2 are as follows: The GMD is omitted; instead, data elements are given for the content type, media type, and carrier type. These are repeatable for different forms of content contained in the resource. For computer-oriented materials, rule 6.9.1.3 gives only the content types “computer program” and “computer dataset.” Video games are considered to be “two-dimensional moving image” or “three-dimensional moving image,” although some catalogers are also adding “computer program.” The same terminology is used for passive or interactive resources (e.g., a CD-ROM consisting of a video file and separate computer program would receive the same terms as a video game). There is not a good way to indicate interactivity or relative degree of media integration by these terms alone. Since games are considered moving images, the source of title is supposed to be the title frames, although optionally one may use the label on the physical carrier (2.2.2.3), giving a note on the source of the title.

The rules for physical description are similar to AACR2, except that computer discs and diskettes are now designated as “computer disc” unless one is exercising the alternative at rule 3.4.1.3 to use a “term in common usage,” a practice that presents the same challenges as it did in 2001.

The systems requirements note is no longer required except in cases “beyond what is normal and obvious for the type of carrier or type of file” (3.20.1.3). There are separate data elements for such information as sound or video characteristics and digital file characteristics (3.16, 3.18–19). However, the terminology for these fields is limited and appears to be optimized for simple resources consisting of one or more files of the same type. Complex resources such as video games are not well described by these rules.

I have concentrated in this section on the physical description of computer games, although AACR2/RDA also covers the formulation of access points. Formulation of access points for titles has proved particularly problematic and will be described in a later section.

DATA FORMAT—MARC

When computer-materials became common enough in library collections for libraries to routinely need to catalog them, it became necessary to create not only cataloging rules, but a data structure that would accommodate those catalog records. The MARC format was already in use for library records, and early records had to be input into existing structures for different types of materials. In 1981, a specific format for machine-readable data files was approved, and it went into widespread use in 1984.²⁰ Fondly known as the MRDF format (pronounced “merdiff”), it was unusual in that it included some fields that were not confined to data described in AACR2. MARC was renamed MARC 21 in 1999.²¹

The MARC bibliographic formats have different fields and byte sequences that are conditional on a byte in the Leader. Leader/06 determines the format of the record, and determines the workform that a cataloger will see in OCLC or other cataloging system when they input a record. Leader/06 “m” is the value for “computer file,” which controls the fields defined for the MRDF format. Most fields are the same as in other formats, but the unique fields include the following:

008/26 Type of computer file. This has a number of different values, such as numeric data, program, representational, document. Interactive multimedia was added as a value at some point, followed by computer games. Early game cataloging frequently used “u” for unknown or “z” for other in this field, other catalogers began using the “i” for interactive multimedia and never switched to “g” for games when it became available.

007 Physical description fixed field. This field is a byte string, with each byte corresponding to a specific value. This field did not become available for electronic resources until the 1990s. The first byte (007/00) is “c” for electronic resource, and the meaning of the rest of the bytes is contingent on that value. Other bytes are type of physical carrier, dimensions, color, and sound. There are eight more bytes, which are seldom used outside of an archival context.

256 Computer file characteristics. This is the MARC field where the AACR2 information for rule 9.3 was recorded (i.e., “computer data,” “computer program(s),” “computer data and program(s)”). It was created in 1987, before that the data was put in the 300 field. Although it is still valid in the MARC format, it has fallen out of use since 9.3 was deprecated in 2001.

753 System details access to computer files. This was designed to be an indexed field, and has subfields for make and model of machine, programming language, and operating system. However, there are no input conventions and no controlled vocabulary for this field, as it is not in either AACR2 nor RDA. Again computer developments quickly outran the instructions, leaving catalogers uncertain of how much detail to include

in this field. Should one input “PC” or “386/33MHz IBM-PC or compatible”? “Game Boy” or “Nintendo Game Boy”? DOS or Windows? With no clear instructions, many catalogers did not use this field, leaving platform information accessible only in the 538 note field. To compensate, many catalogers bend the rules for edition and put the game platform in the 250 Edition field, on the theory that it designates a change from another version—even when there is not another version. There are also numerous note fields, numbered 500 to 588, some of which are specific to computer resources but many of which are seldom used because they relate to data sets. Field 538 is used for the Systems requirements note, and though the cataloging rules called for this note to be placed first, in many systems the notes displayed in numerical order, placing it far down in the record. Some catalogers also use this field for a “Disc characteristics” note to record that a disc is, for example, a CD-ROM or burnable disc, but this note is not called for in the cataloging rules. The 500 note, the “general” note, became the dumping ground for much useful information.

Since the approval of RDA, fields were hastily added to the MARC 21 format to enable the encoding of, first, the content type (336), media type (337), and carrier type (338). Later, fields were added for sound characteristics (344), video characteristics including video systems NTSC and PAL (346), plus field 347 for digital file characteristics with subfields for file type, file encoding format, file size, resolution, regional encoding, and transmission speed. While laudable to finally have fields enabling more granular encoding of technical details, the caveats given in the above discussion of RDA rules 3.16, 3.18–19 apply.

SUBJECT-GENRE

Subject and genre headings have historically been intertwined, so it is best to deal with them together.

American libraries generally use *Library of Congress Subject Headings* (LCSH).²² These are topical headings, supplemented by headings from the online Library of Congress Name Authority File (NAF) when the names of people or the titles of works are needed as subjects. Individual catalogers may propose subject headings, but their approval is under the editorial control of the Library of Congress (LC). Formerly LC cataloged games according to AACR2. However, they are now within the moving image archives division (currently called Motion Picture Broadcasting and Recorded Sound Division or MBRS), which does not use AACR2 for cataloging.²³ Thus, game cataloging at LC no longer falls under the purview of the rules and practices of the Policy and Standards Division, so there is no constituency within that division with a need to make and revise policies for game cataloging. This

leaves a huge policy gap, as most American libraries follow the LC policies for all materials, even the ones that LC does not catalog.

Assigning subject headings may or may not be straightforward. There are some games for which a topic is apparent, such as games re-creating World War 2 battles or Roman civilization, or games involving sports or sport leagues. Other topics are not so clear. What is *The Sims* about? People? Is the subject heading "Persons" useful for this material? Is *Earth Worm Jim* really about "Earthworms"? How about fantasy games in imaginary places? "Imaginary places"? Various fight and first person shooter games? "Hand to hand fighting"? "Shooting"? It is not clear how useful these topics are, and in any case, assigning subject headings is a notoriously subjective art, and many catalogers provide minimal subject headings for computer games.

There are two methods of providing genre information in a catalog record in an indexed field. The most straightforward is to add a form/genre subdivision to the subject heading. Form/genre subdivisions have been in use for some time, though they were not explicitly given a subdivision tag (subfield "v") in the MARC format to distinguish them from topical subdivisions until the late 1990s. There is a form subdivision –Computer games which may be added to a subject such as the ones above, leading to possibly helpful heading such as *Ardennes, Battle of the, 1944–1945—Computer games* or possibly unhelpful ones such as *Hedgehogs—Computer games*. There is also a form/genre subdivision –Interactive multimedia, which some catalogers use for games. There is a subdivision for –Simulation games but it is explicitly defined in LCSH as topical, and not a form or genre, as is –Computer simulation. The subdivision –Software was authorized in the 1990s, prior to the authorization of –Computer games. A workshop given by the LC in San Francisco in 1997 had –Software as the recommended form subdivision for games in the printed handouts, but attendees were given verbal instructions to use the new subdivision –Computer games, as they had not intended the subdivision –Software to apply to all materials in electronic format, just those which were executable programs.²⁴ However, many catalogers still use the subdivision –Software, often after the names of video game genres. The example *Computer adventure games—Software* is still given in the *Subject Headings Manual*, H 2070 "Software and works about software."²⁵ No explanation is given, making it unclear if it is supposed to be referring to a resource that is a computer adventure game, or to a software program to create a computer adventure game.

LC does have headings for game genres. However, they have all been established as topical headings rather than genre headings. These headings were established for the purpose of providing subject headings for books about particular game genres, and are established on an as-needed basis. They do not reflect the actual universe of game genres. In the 1990s before games were moved to MBRS, they sometimes used a topical game genre heading as a genre heading, although they still gave it the MARC tagging

for a subject. It was not until around 2000 that LC began using the MARC tag 655 for genre headings, and a few years further on until LC began distributing authority records specifically designated as genre headings. They have distributed headings for film and television genres, radio, maps but as yet have no video game genre headings in the genre authority file. Most libraries use the limited video game genre headings, but coded as topics (MARC field 650).

Libraries are free to input genre headings coded as genres (MARC 655), but the LC subject headings list is the only list in widespread use. Even though there are no game headings in the genre list, there is a little-known workaround for libraries. The Frequently Asked Questions About Library of Congress Genre/Form²⁶ says that libraries submitting official BIBCO records for the Program for Cooperative Cataloging may use an LC subject heading as a genre tagged 655 #0 if the heading does not have a scope note saying that it is only “about” or “on” works of a given type. Otherwise, if they decide to use the heading it must be tagged as 650 #7, indicating it is a “local” heading.

According to these instructions, the heading “Computer games” may not be used as a genre because it has a scope note specifying it as a topic, but the following headings lack scope notes so may be coded as genres in the records for individual games:

- Video games
- Computer adventure games
- Computer baseball games
- Computer chess
- Computer flight games
- Computer war games
- Computer word games
- Video keno
- Video poker
- Video wrestling games
- Educational games
- Simulation games
- Fantasy games (this is a narrower term from Role playing)

LC also has headings for specific brands of games, such as Sega Dreamcast video games, but not for all brands. Again, these were established because a book was being cataloged that was about a particular brand of games. Many catalogers also use these as genre or topical headings on the records for individual video games, although it is not clear that this is permissible or useful. Platform and brand information does not seem to be a genre by any stretch of the imagination.

Note that the authorized genre subdivision for a topic is —Computer games. Note that the term “Computer games” is not authorized by itself as a genre, although various types of computer games are able to be used by

the workaround noted above, as is the term Video games. In cataloging a single game, most catalogers are currently using the topic —Computer games construction when possible, and adding one of the more specific genres, like Computer war games. If the game is played on a computer, they stop there. If it is played on a video game device such as Xbox or Game Boy or PlayStation, they also add the heading Video games.

This begs the question: Is there a difference between computer games and video games? LC seems to think so, since both headings exist in the LCSH list, with individual game titles given as narrower terms, and the lists do not overlap (see the next section on titles for more details about the game titles in the subject file). What constitutes a video game and what constitutes a computer game is not defined. Catalogers generally suppose that it means that if a game is played on a computer it is a computer game and if it is played on a dedicated video game device it is a video game. However, since the only permissible game subdivision is —Computer games and catalogers often use a mix of “computer” and “video” game headings on the same records, it is not clear at all if this distinction is meaningful or helpful. Should all be “Video games”? There is already a LCSH for “Electronic games,” which is a broader term to both Video games and Computer games. Should that be used instead? It has a scope note that forbids its use as a genre. And there are electronic games that are not video games.

Clearly the entire field of game genre headings and subdivisions needs to be completely overhauled.

A SPECIAL NOTE ON TITLES

In library cataloging, the title of the resource may serve different functions in a bibliographic record. The title is, of course, the title of the resource the cataloger is holding in hand. However, if a resource were previously issued under another title or in a different language, there is a need to bring together all versions of the title so that all versions may be found by the catalog user. An additional title is used for this, called a “uniform title” under AACR2 and the “preferred title” under RDA. For materials with one or two creators, this is formulated by using author’s name (last name first) followed by a period and the original title of the resource. For materials of diffuse creation, including most modern games, the title alone is used. In addition to the collocation function, this uniform title may also be used as a subject heading for works which are about the work (e.g., a strategy book for a particular video game). The use of the uniform title was considered optional under AACR2 except for libraries following the LC guidelines, particularly libraries contributing Bibliographic Record Cooperative Program (BIBCO) records as part of the Program for Cooperative Cataloging. As the uniform title might be identical to the title of the item being cataloged, there were only certain circumstances

when one would make an additional entry for this uniform title, and still more restricted circumstances when one would contribute a record to the LC NAF. Therefore, the creation of a name authority record seldom occurred when one was cataloging the game itself.

When LC first needed to use the titles of computer resources, they were generally cataloging books about a computer program rather than the program itself. Therefore, their rules called for establishing the titles as subject headings. Titles which were established in the subject file were specifically coded as topical subject headings, and could therefore not be used as name headings. When catalogers began cataloging the computer resources themselves, they needed to use name headings as names, not topical subjects. Since 1987, the LC documentation instructs their catalogers to establish the headings for "individual computer programs . . . when needed, in the name authority file."²⁷ When cataloging a work about a program, catalogers are to search the LC database to see whether a name authority record or bibliographic record has been created already and use the form of title found there, but code it in their bibliographic record as a title subject (MARC 630) rather than a topical subject. If the program has not been cataloged by LC, they are to have someone in the Special Materials Cataloging Division establish a heading in the NAF. If a conflict is found with something else with the same title, the qualifier (Computer file) is added to the title. No instructions are given for the case where the heading is already established as a topical subject (MARC 650), therefore not useable as a title subject.

When cataloging a computer resource itself, no specific instructions were given for establishing a "uniform title" aside from the general rules for all materials. As the title would only be needed in the rare instances when it was previously issued under another title or in a different language, the issue rarely came up (and when it did, it was sometimes because a game had been issued previously in Japan and likely the cataloger could not determine the authorized Japanese form in order to use the original title). In 1987, when the LC was to cease establishing the titles of computer programs as subjects, they added a rule interpretation to 25.5B stating that if a computer program were entered under the same title as the title work, do not assign a uniform title simply to distinguish, but if the uniform title were otherwise needed, use the qualifier (Computer program), and then use that uniform title, with the qualifier, for all entries for that particular work—including changing other bibliographic records for that work.²⁸ Like similar rules for motion pictures and television programs, this was widely ignored by the cataloging community, and apparently by LC as well since it is contradicted by the aforementioned policy in the Subject Cataloging Manual.²⁹ Some minor clarifications were made over the years, but the rule interpretation was deleted with Cataloging Service Bulletin no. 99 (Winter 2003).

Note that in both the Subject cataloging rules and in the LC rule interpretations, the subject under discussion was “computer programs.” Computer or video games are not mentioned in the discussion of title practice. Does this mean that these rules did not cover games? Most catalogers assumed that the rules were intended to cover all computer materials, a feeling reinforced when in the late 1990s the rule interpretation for computer programs was expanded to cover electronic serials, which were clearly not “programs.”

With the adoption of RDA, the LC in its Policy Statement for 6.27.1.9 of June 2012 declared that for monographically issued resources the cataloger must add a preferred title with a qualifier in case of a conflict.³⁰ It gives general rules for all types of materials but no specific clues for computer resources or games. It offers up the choice of corporate body, date of publication, “descriptive data elements” (e.g., edition statement), place of publication, or if none of those are appropriate, “any word(s) that will serve to distinguish the one work from the other,” and prescribes the qualifiers (Motion picture), (Television program), (Comic strip), (Choreographic work), and (Radio program) for those types of works. With the slow adoption of RDA, many catalogers have not yet seen nor realized the implications of this policy change, and in any case they are binding only on the LC, which no longer catalogs computer games in their library workflow, and libraries contributing BIBCO records to the Program for Cooperative Cataloging.

To summarize the labyrinthine situation, catalogers for video games who are following the Library of Congress policies for preferred title must check at least the LC NAF and bibliographic file, as well as any other cooperative catalog (such as OCLC) with which the cataloger works, to determine whether the title of the game being cataloged is the same as that of another work which is also entered under title. If a work with the same title is found, it is considered a conflict. If a game is a franchise related to a television program or motion picture, it will almost surely be a conflict. It will also conflict if a cataloger chooses to give the franchise title as the title and the distinctive game titles as a subtitle (a common, but problematic practice). The title also must not conflict with an LC subject heading, a not-infrequent occurrence with one-word titles. If a conflict is found, the cataloger must add a qualifier to the title, but no instructions are given for video games. Should the cataloger follow the rules for monographs (publisher, date, etc.) or follow the pattern for movies and television programs? The latter is more user-friendly but in the absence of a specific directive, one assumes that the general rule for monographs should be followed. However, looking back at the Subject Cataloging manual rule for H2070, which has not been superseded so is still in force, it still says to use the qualifier (Computer file) when the title is needed as a title subject, even though it is to be established in the name file. This is assuming that games are supposed to be covered under that rule, which is not entirely clear. Catalogers (in the rare instances that they are establishing these titles in the name authority file) clearly do not

know what qualifier to use, as a search through the name authority file will show headings using (Computer game), (Video game), (Game), and so on. To further muddy the waters, even though names of “computer programs” are supposed to have been established in the NAF since 1987, new game titles are continuing to be added to the subject file, for example, “Angry Birds (Game),” added in April 2012 and “Angry Birds: Star Wars (Game),” added in October 2012. The qualifier (Game) is being given to these titles in the subject file, the same as is used for a traditional or board game, though there are no explicit rules in the subject cataloging manuals for games of any type. There are no instructions for what to do when the title would conflict with the name of another type of game such as Monopoly (Game) or Solitaire (Game)—though there is also a subject heading for Solitaire (Board game).

CONCLUSION

Video game cataloging has undergone a great evolution since the general appearance of catalogable computer resources in the late 1970s. Cataloging rules and the MARC 21 have been frequently updated as the characteristics of the materials changed, usually lagging behind by a few years. Consistent terminology has been elusive, making it difficult to collocate materials or provide a consistent search or limit strategy to find all like materials. RDA and the recent MARC 21 updates have provided more granularity and some controlled vocabulary, but continue to better describe simple electronic resources and data files than complex media like games and other interactive works.

In contrast to the somewhat improving situation regarding the descriptive cataloging rules and MARC, the policies for subjects, genres, and uniform/preferred titles remain intractable. The LC sets the policies that most American libraries follow for uniform/preferred titles of works and for subject and genres headings. But as they no longer catalog games within the library section of their operations, their policies are inadequate and contradictory for these materials. They have had neither the mandate nor the staffing to undertake the task of completely reevaluating their policies on game genres headings, subject headings, and name heading policies to bring some consistency and logic to them.

In the meantime, catalogers continue reading the tea leaves to try to figure out how they are to deal with these complex, yet ubiquitous materials.

FUNDING

This research was supported by IMLS Grant LG-06-13-0205-13.

NOTES

1. American Library Association, *A.L.A. Cataloging Rules for Author and Title Entries* (Chicago: American Library Association, 1941).
2. American Library Association, *Anglo-American Cataloging Rules* (Chicago: American Library Association, 1967)
3. American Library Association, Canadian Library Association, and Library of Congress, *Anglo-American Cataloging Rules: North American Text: Chapter 12 Revised, Audio-Visual Media and Special Instructional Materials* (Chicago: American Library Association, 1975), v–vi.
4. Michael Gorman and Paul W. Winkler, *Anglo-American Cataloguing Rules* (Chicago: American Library Association, 1978), 202–203.
5. Sue A. Dodd, *Cataloging Machine-Readable Data Files: An Interpretive Manual* (Chicago: American Library Association, 1982), 120.
6. Nancy B. Olson, *A Manual of AACR2 Examples for Microcomputer Software and Video Games* (Lake Crystal, MN: Soldier Creek Press, 1983), 13.
7. *Ibid.*, 19.
8. *Ibid.*, 21. In corrections to the examples issued in 1984, the statement was amended to read “8 program files.”
9. American Library Association, *Guidelines for Using AACR2, Chapter 9 for Cataloging Microcomputer Software* (Chicago: American Library Association, 1984), 1–2.
10. *Ibid.*, 9–10.
11. *Ibid.*, 5–6.
12. Sue A. Dodd and Ann M. Sandberg-Fox, *Cataloging Microcomputer Files: A Manual of Interpretation for AACR2* (Chicago: American Library Association, 1985), xi.
13. Dodd and Sandberg-Fox, *Cataloging Microcomputer Files*, xii.
14. Gorman and Winkler, *Anglo-American Cataloguing Rules*, 221.
15. Association for Library Collections & Technical Services and Laurel Jizba, *Guidelines for Bibliographic Description of Interactive Multimedia* (Chicago: American Library Association, 1994), 1.
16. Michael Gorman and Paul W. Winkler, *Anglo-American Cataloguing Rules. Amendments 2001* (Ottawa: Canadian Library Association, 2001).
17. Joint Steering Committee for Revision of AACR and American Library Association, *Anglo-American Cataloguing Rules* (Ottawa: Canadian Library Association, 2002).
18. The Joint Steering Committee for Development of RDA, *RDA: Resource Description and Access* (Chicago: American Library Association, 2011).
19. RDA is closely tied with The Functional Requirements for Bibliographic Records (FRBR). For a discussion of the difficulty of applying FRBR to video games as well as the more mundane problems of sourcing information on video game metadata properties, see Jin Ha Lee, Joseph T. Tennis, Rachel Ivy Clarke, and Michael Carpenter, “Developing a Video Game Metadata Schema for the Seattle Interactive Media Museum,” *International Journal on Digital Libraries* 13, no. 2 (20130307): 105–117.
20. Dodd and Sandberg-Fox, *Cataloging Microcomputer Files*, 3–4.
21. Library of Congress (Washington), and Bibliothèque nationale (Canada), *MARC 21 Format for Bibliographic Data: Including Guidelines for Content Designation* (Washington, DC: Library of Congress, 1999)
22. Library of Congress, *Subject Headings Manual* (Washington, DC: Library of Congress, Cataloging Distribution Service, 2011), H 2070 “Software and Works about Software,” August 2008 update, 1.
23. Trevor Owens, “Yes, The Library of Congress Has Video Games,” September 26, 2012. <http://blogs.loc.gov/digitalpreservation/2012/09/yes-the-library-of-congress-has-video-games-an-interview-with-david-gibson/>
24. Lynn El-Hoshy and Dave Reser, “Demystifying Subject Cataloging of Electronic Resources,” ALCTS Preconference, Pan Pacific Hotel, San Francisco, California, June 26–27, 1997. Comment on verbal instructions based on notes by the author.
25. Library of Congress, *Subject Headings Manual*, H 2070, 1.
26. Library of Congress, “Frequently asked Questions about Library of Congress Genre/Form Terms for Library and Archival Materials (LCGFT),” revised June 6, 2011. http://www.loc.gov/catdir/cpsd/genre_form_faq.pdf, question 35. An earlier version of this document

further specified that you could use headings if LC had itself used a heading as a genre heading, whether they had coded it 650 or 655. Since LC had not cataloged any games since the 1990s, the pool of LC records to check for examples was limited.

27. Library of Congress, *Subject Headings Manual*, H 2070, 2.

28. Robert M. Hiatt, *Cataloging Service Bulletin* (Washington, DC: Library of Congress, Processing Services, 1978), 38 (Fall 1987): 48.

29. Ibid.

30. Library of Congress, *Library of Congress-Program for Cooperative Cataloging Policy Statements (LC-PCC PS)*. Rule 6.27.1.9, section on monographs, version accessed was last revised December 2012.

APPENDIX: EXAMPLES OF CATALOG RECORDS FOR VIDEO GAMES, 1983–2011

From Nancy B. Olson, *A Manual of AACR2 Examples for Microcomputer Software and Video Games* (Lake Crystal, MN: Soldier Creek Press, 1983), page E 41, example retyped but otherwise unchanged. In the corrections issued in January 1984 the number 8 was added to the number of program files. Used with permission from the author.

PAC-MAN [game]. – Chicago, IL : Sears, Roebuck and Co., c1981

program files in 1 computer cartridge
(Atari 2600) + game instructions. – (Sears
tele-games)

Title from cartridge label.

System requirements: Atari 2600 with monitor,
joystick controllers.

Summary: Eight versions of a video game
for 1 or 2 players. To keep PAC-MAN happy,
you must guide him around the maze to eat
video wafers and power pills, protect him
from ghosts, and help him gobble up ghosts
before they gobble him.

“video arcade.”

From Sue A. Dodd and Ann M. Sandberg-Fox, *Cataloging Microcomputer Files: A Manual of Interpretation for AACR2* (Chicago: American Library Association, 1985), page 248, example retyped but otherwise unchanged. Used with permission from the American Library Association.

Computer baseball strategy [machine-readable data file] /
[computer game design, Ross Maker; computer game
programmer, Stephen Goss; computer graphics

animation, Daniel Nicholson]. — Baltimore, Md. : Microcomputer Games, c1982.

4 program files + 1 instruction booklet. — (Sports illustrated)

System requirements: 32K; Apple II and II+ with Applesoft BASIC in ROM; Atari 400/800; Commodore Pet/CBM with 40 character wide display screen; TRS-80 Models I & III, Level II BASIC.

In: 1 computer cassette.

Title from cassette label.

In container (29 cm.)

Summary: Four machine versions of a baseball game that allows the player to act as manager of the team.

1. Baseball. 2. Games. I. Maker, Ross. II. Goss, Stephen. III. Nicholson, Daniel. IV. Series.

From Deanne Holzberlein, "Computer software cataloging: techniques and examples," *Cataloging & Classification Quarterly*, 6, no. 2 (Winter 1985–86), page 57; published by Haworth Press. Example retyped but otherwise unchanged. Reprinted by permission of the publisher (Taylor & Francis Ltd, <http://www.tandfonline.com>) and the author.

Sherman, Larry.

Crossword magic [machine-readable data file] / by Larry Sherman. — Version 3.0. — Sunnyvale, CA (P.O. Box 70728, Sunnyvale, CA 94086) : L&S Computerware, c1981.

1 program file on 1 computer disk : double sides, col. ; 5 1/4 in. in container, 13 × 16 × 1 cm. + 1 guide ([4] p.; 22 cm.).

System requirements: Apple II+; 48K; DOS 3.3.

Summary: For constructing British non-symmetrical puzzles.

Contains: Maker disk — Demonstration disk.

1. Crossword puzzles—Computer programs. I. L & S Computerware. II. Apple II—DOS 3.3. III. Title.

LC Control No.: 91735055
 LCCN Permalink: <http://lccn.loc.gov/91735055>

000 01678nmm a2200397 a 450
 001 4998949
 005 19930628105554.3
 008 910403s1986 mdu b
 035 __ |9 (DLC) 91735055
 906 __ |a 7 |b cbc |c orignew |d u |e ncip |f 19 |g ygencompf
 955 __ |a vf03 B&E 06-15-93; HQ to PSSA 06-17-93
 010 __ |a 91735055
 020 __ |a 0872760049
 040 __ |a DLC |c DLC
 041 0_ |g eng
 050 00 |a HQ25
 082 00 |a 306.7 |2 12
 245 00 |a Dr. Ruth's computer game of good sex |h [computer file].
 256 __ |a Computer program.
 260 __ |a Baltimore, Md. : |b Avalon Hill Game Co., |c c1986.
 300 __ |a 1 computer disk ; |c 5 1/4 in. + |e 1 booklet.
 538 __ |a System requirements: IBM PC and MSDOS compatibles; 128K.
 500 __ |a Title from disk label.
 500 __ |a Project supervision, Ruth Westheimer; game design, Michael E. Moore, Mark Herman; IBM programming, Bob Gardner.
 500 __ |a Based on the board game by Victory Games, Inc.
 521 __ |a Adult players.
 530 __ |a Issued also for Apple and Commodore 64 computers.
 520 __ |a A fun and informative game for one to seven people. Each player gets to answer up to eight questions about love, sexuality, and relationships; exceptional play leads to a bonus question from the Sex Clinic. Players with the two highest scores faceoff in a head-to-head contest.
 650 _0 |a Sex |x Miscellanea |x Software.
 650 _0 |a Computer games |x Software.
 700 1_ |a Westheimer, Ruth K. |q (Ruth Karola), |d 1928
 710 2_ |a Avalon Hill Game Company.
 740 0_ |a Doctor Ruth's computer game of goodsex.
 740 0_ |a Good sex.
 753 __ |a IBM PC
 991 __ |b c-MRC |h HQ25 |t Copy 1 |w CF

FIGURE A1 Example Cataloged by the Library of Congress Which Appeared in Shorter Form as an Example in the Handouts for Lynn El-Hoshy and Dave Reser, “Demystifying Subject Cataloging of Electronic Resources” : ALCTS Preconference, Pan Pacific Hotel, San Francisco, California, June 26–27, 1997. This Version Appears to Have Been Cataloged in Approximately 1991; This View Is Taken from the MARC Display in the Library of Congress Online Catalog, March 7, 2014.

LC Control No.: 99382150
 LCCN Permalink: <http://lcn.loc.gov/99382150>
 000 01734nmm a2200349 a 450
 001 5006825
 005 19990128112123.6
 007 co|||
 008 990128s1993 cau g eng
 035 __ |9 (DLC) 99382150
 906 __ |a 7 |b cbc |c orignew |d u |e ncip |f 19 |g ygencompf
 955 __ |a vb22 to bccd 01-28-99
 010 __ |a 99382150
 040 __ |a DLC |c DLC
 050 00 |a GV1469.22
 082 10 |a 793.93 |2 13
 245 00 |a LucasArts presents Sam & Max hit the road |h [computer file].
 246 30 |a Sam & Max hit the road
 246 30 |a Hit the road
 246 3_ |a LucasArts presents Sam and Max hit the road
 256 __ |a Computer data and program.
 260 __ |a [San Rafael, Calif.] : |b LucasArts, |c c1993.
 300 __ |a 1 computer laser optical disc ; |c 4 3/4 in.+ |e 1 hintbook + 1 reference card + 1 coloring book.
 538 __ |a System requirements: 386/33MHz IBMcompatible PC; 4MB RAM; MSDOS 5.0 or higher; MSCDEX 2.1 or better; 256color VGA graphics display; MPC level 1-compatible CD-ROM drive; Sound Blaster or compatible audio card; keyboard, mouse, or joystick. Hard disk drive required for saved games.
 500 __ |a Title from disc label.
 500 __ |a Created by Steve Purcell.
 500 __ |a Issued also for Macintosh.
 520 __ |a Follows the humorous adventures of Sam andMax, a canine and rabbit freelance police team. They explore strange roadside attractions throughout America in their search for Bruno, the runaway carnival bigfoot. Features animation, music, and popup verb and dialog icons. Music may also be played on compact disc player.
 650 _0 |a Computer adventure games |x Software.
 700 1_ |a Purcell, Steve.
 710 2_ |a LucasArts (Firm)
 991 __ |b c-MRC |h GV1469.22 |i [1999 00011] |t Copy 1 |w CF

FIGURE A2 Another Example of a Game Cataloged by the Library of Congress, This One in Approximately 1999. This View Is Taken from the MARC Display in the Library of Congress Online Catalog, March 7, 2014.

LEADER 01194cmm a22002774a 4500
 001 a5639904
 003 SIRSI
 005 20140208050005.0
 007 co cga| || || || || ||
 008 040520s2003 xxu geng d
 035 a| (OCoLC-M)79044141
 035 a| (OCoLC-I)275843030
 037 a| SLUS 20666 b| Atlus
 040 a| CSt c| CSt
 245 00 a| Disgaea h| [electronic resource] : b| hour of darkness.
 246 30 a| Hour of darkness
 260 a| Irvine, CA : b| Atlus USA Inc., c| c2003.
 300 a| 1 CD-ROM : b| sd., col. ; c| 4 3/4 in. + e| 1 game manual (24 p. ; 19 cm.)
 500 a| Title from disc label.
 520 a| Game "follows the misadventures of the young prince Larharl and his rag-tag army as the
 bettle demons and foul creatures forsupremacy over the Netherworld."
 538 a| System requirements: PlayStation 2.
 596 a| 1
 650 0 a| Computer adventure games v| Software. =| ^A2097169
 710 2 a| Atlus U.S.A., Inc. =| ^A1948869

FIGURE A3 Example from the Stanford University Online Catalog, Cataloged in 2005. In the Current OCLC Version of this Record (as of March 7, 2014), Someone Has Added the 655 0 Sony Video Games.

Rec stat c	Entered 20110426	Replaced 20110824			
Type m	ELvl	Src c	Audn	Ctrl	Lang eng
BLvl m		G Pub		MRec	Ctry cau
Desc i	File g		DtSt t	Dates 2011	, 2011
007	c #b o #d c #e g #f a				
040	___ #b eng #c ___ #e rda				
024 1	014633157963				
042	pcc				
245 0 0	Crysis 2 / #c Crytek.				
246 3	Crysis two				
250	Limited edition.				
264 1	Redwood City, CA : #b Electronic Arts, #c [2011]				
264 4	#c ©2011				
300	1 computer disc: #b DVD-ROM, sound, color; #c 4 3/4 in. + #e 1 booklet(13 pages: illustrations; 19 cm)				
336	two-dimensional moving image #2 rdacontent				
336	computer program #2 rdacontent				
337	computer #2 rdamedia				
338	computer disc #2 rdacarrier				
380	Video games.				
538	System requirements: Windows XP (SP3), Windows Vista (SP1 or Windows 7; 2 GHz Intel Core 2 Duo or 2 GHz AMD Athlon 64 X2 or better; 2 GB or more RAM (2 GB for Windows Vista (SP1)); 8x DVD drive or faster; at least 9 GB of free hard disc space; DirectX 9.0c compatible video card (512 MB, details of supported video cards on container); DirectX 9.0c compatible sound card; may be controlled by keyboard, mouse, or Mixrosoft XBox 360 Controller for Windows.				
500	First-person shooter computer game, 3-D compatible.				
500	Title from disc label.				
521 8	ESRB rating: M, Mature 17+ (Blood, partial nudity, strong language, violence)				
520	"In the role of a soldier, alone in a devastated urban jungle, players use the super-human abilities provided by a technologically advanced suit to save the world from alien invaders, as they battle through the first-person shooter sequel, Crysis 2. As in the first game, the hero's most important weapon is a high-tech suit that integrates with his body to provide increased defense, enhanced stealth, and various offensive capabilities. The sequel's enemy aliens employ improved situational intelligence and group tactics, calling on players to approach each situation strategically and make the most of the array of powers provided by the suit. Written by science fiction author Richard Morgan, the story takes place in an embattled New York City. The metropolis crumbles under a full-on invasion, as aliens return in force to annihilate humankind. Conventional weapons and warfare fail against the onslaught, leaving Earth's salvation to the wearer of the revolutionary Nanosuit 2. Using the integrated stealth, strength, and special abilities provided by the futuristic armor -- along with a selection of highly destructive weapons -- players embark on dangerous missions to strike at crucial vulnerabilities and sabotage the alien assault. The game's multiplayer options include "Crash Site," a team-based mode in which players compete for control of resources that appear regularly on the battlefield, and "Team Instant Action," for pick-up-and-play, squad-based deathmatches."-- All game guide/				
650 0	<u>Combat</u> #v <u>Computer games</u> .				
655 0	Computer war games.				
700 1	Morgan, Richard K., #d 1965-				
710 2	<u>Crytek (Firm)</u>				
710 2	<u>Electronic Arts (Firm)</u>				

FIGURE A4 Example from PCC SCT RDA Record Examples, Downloaded from http://www.loc.gov/catworkshop/RDA%20training%20materials/SCT%20RDA%20Records%20TG/Records/16_Video%20Games_Final.docx on March 7, 2014. This Example Was Published before the Implementation of the MARC21 34x Fields.